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# The effects of digital filtering on mismatch negativity in wakefulness and slow-wave sleep

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**SUMMARY** The mismatch negativity (MMN) is a response to a deviant auditory stimulus that occurs infrequently in a sequence of otherwise repetitive, homogeneous standard auditory stimuli. The MMN is presumed automatic and independent of conscious awareness. Recording of the MMN during unconscious states may be problematic. The frequency content of the long-lasting MMN may overlap and summate with other event-related slow potentials and low-frequency background electroencephalogram (EEG) activity. The purpose of this study is to determine the optimal filter settings for recording the MMN during unconscious states. Auditory event-related potentials (ERPs) were recorded from eight subjects in an oddball paradigm during wakefulness and Stages 3 and 4 of sleep [slow-wave sleep (SWS)] using a 0.16–35 Hz analogue bandpass. Deviant probability was 0.033. Stimulus-onset asynchrony was 150 ms. The EEG data were subsequently digitally filtered in the frequency domain. The low-pass filter was set at either 24, 12 or 6 Hz, and the high-pass filter at either 1, 2, 3 or 4 Hz. Applying a low-pass filter down to 12 Hz had a minimal impact on the waking or sleeping MMN amplitude. On the other hand, increasing the high-pass setting from 2 to 3 Hz permitted the visualization of the MMN recorded during sleep. The 4 Hz filter showed a similar trend but also markedly attenuated the amplitude of the waking MMN. A high-pass setting of 3 Hz provides a reasonable compromise. It has only a slight effect on the MMN when the subject is conscious, but still attenuates most of the unwanted slow potential activity when the subject enters SWS.

KEYWORDS event-related potentials, filter settings, mismatch negativity, unconsciousness, slow-wave sleep

## INTRODUCTION

When a subject is presented with a deviant auditory stimulus occurring infrequently in a sequence of otherwise repetitive, homogeneous standard auditory stimuli, a mismatch negativity (MMN) is elicited (Näätänen *et al.* 1978). The MMN is a long-lasting, low-amplitude negative event-related potential (ERP) wave that overlaps in time and space with the N1–P2 vertex potential. The MMN can begin as early as 50 ms and last up to 300 ms after deviant onset. The MMN has come to serve as an index of entry into or out of consciousness states (e.g. sleep, coma, general anesthesia). However, recording the MMN during unconscious states presents a series of problems that are not encountered in the waking state.

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During wakefulness, Campbell *et al.* (1992) noted that the auditory N1–P2 waveform is overlapped by another long-lasting waveform, the waking processing negativity (wPN). Unlike the MMN, the wPN occurs following both the standard and deviant stimuli. Moreover, it occurs even in subjects who are apparently ignoring the auditory stimuli. Sensory input is markedly inhibited during unconscious states. This is associated with the removal of the overlapping PN. As such, N1 declines in amplitude to near baseline level while P2 may appear to increase in amplitude. The gradual positive DC shift that occurs in unconscious states may summate and overlap with the MMN that occurs to the deviant stimulus, making it difficult to observe.

A means to reduce the amplitude of the slow-wave activity is through judicious use of high-pass filtering. The frequency spectrum of the long-lasting MMN may, however, partially overlap with that of the low-frequency slow-wave activity. Some compromise must be exercised in order to avoid filtering the signal of interest, the MMN, while still attenuating the background, overlapping noise.

The MMN gradually becomes attenuated from the waking to sleeping state (Nittono *et al.* 2001; Sabri *et al.* 2000). During SWS, the majority of studies have failed to observe an MMN following either frequency (Nielson-Bohlman *et al.* 1991; Paavilainen *et al.* 1987; Sallinen *et al.* 1994; Winter *et al.* 1995) or intensity (Loewy *et al.* 2000) deviants. It is possible that the failure to observe the MMN is because of the high signal-to-noise ratio common to SWS. The amplitude of the MMN is very small (often less than 1 $\mu$V) and the amplitude of the background delta electroencephalogram (EEG) is very large (often more than 200 $\mu$V). To reduce background noise sufficiently, many deviant stimuli need to be presented and averaged together. Unfortunately, because of the relatively limited duration of SWS, this has not been possible.

During the waking state, a bandpass ranging from 0.1 to 35 Hz has often been used to record the MMN. During unconscious states various high-pass filters ranging from 0.1/0.3 (Atienza *et al.* 1997, 2000; Nordby *et al.* 1996; Sabri *et al.* 2000) to 1.0 Hz (Kane *et al.* 1996; Sallinen *et al.* 1994; Sallinen and Lyytinen 1997) have been employed. Still others increase the high-pass setting to 3 Hz (Fischer *et al.* 1999; Loewy *et al.* 1996, 2000) while others decrease it to 0.01 Hz (Nittono *et al.* 2001). Larger high-pass settings (near 4 Hz) should serve to reduce the amplitude of the overlapping slow-wave activity of the processing negativity. However, these filter settings may also reduce the amplitude of the true MMN. The purpose of the present study was to determine the systematic effects of varying both the low- and high-pass filter settings on the MMN recorded during the waking state and within SWS.

## METHODS

Eight (five men, three women) healthy young adults (aged 21–42, mean = 26 years) volunteered for this study. The data were collected during a single session in the sleep laboratory. The EEG was recorded from four scalp locations placed at midline frontal (Fz), central (Cz), parietal (Pz), and occipital (Oz) sites, and from the right mastoid (RM). The reference was the nose. A vertical and horizontal electrooculogram (EOG) were also recorded. Inter-electrode impedances were kept below 5 k$\Omega$. Data were acquired continuously using a 256-Hz sampling rate and later reconstructed into discrete 500 ms 'sweeps'. The analogue high- and low-pass filters were set at 0.16 and 35 Hz (3 dB attenuation), respectively.

Evoked potentials were elicited using standard 1000 and deviant 2000 Hz tone pips (80 dB SPL; 50 ms duration; rise/fall time of 5 ms). The extent of deviance was thus very large compared with that employed in most waking studies because the MMN is best elicited by very discrepant deviants during sleep (Atienza *et al.* 1997; Loewy *et al.* 1996). Probability of occurrence of the standard and deviant stimuli was 0.967 and 0.033, respectively. Standard and deviant stimuli occurred pseudorandomly with the restriction that two deviants could not be presented consecutively. Stimuli were presented at a constant stimulus onset asynchrony of 150 ms. A total of 5000 trials was presented per block. During wakefulness, subjects ignored the auditory stimuli by reading a self-selected book or magazine. Four blocks of stimuli were presented in the waking state. A minimum of 10 blocks of stimuli were presented when subjects were sleeping. If subjects awoke during the testing session, stimulus presentation was halted and resumed 5 min after definitive sleep onset.

During wakefulness, trials in which the EEG or EOG exceeded $\pm 100$ $\mu$V were rejected from further analysis. During sleep, the artifact reject was set to $\pm 150$ $\mu$V to accommodate the higher amplitude EEG. The different stages of sleep in each block were classified according to the standardized criteria of Rechtschaffen and Kales (1968). In cases of stage ambiguity, the epoch was excluded from further analysis. For purposes of this study, only data from the waking state and SWS were compared. Separate averages were computed for the standard and deviant stimuli. The MMN is best observed in a subtraction waveform. The different waveforms were computed by subtracting, point-by-point, the standard from the deviant waveforms at each electrode site within wakefulness and SWS.

Event-related potentials were subsequently digitally filtered (3 dB attenuation) in the frequency domain using a technique developed by Yuen and Fraser (1979). ERPs were fast-Fourier transformed (FFT) and then multiplied by a transfer function. The roll-off of the stop-bands was defined by a half cosine function. In order to prevent box-car filter shapes, the original ERPs were padded with zeroes to increase resolution. Finally, the filtered ERPs were inversely transformed (i.e. the inverse of FFT) to the time domain. Frequency-based digital filtering has the advantage of avoiding the types of phase shifts that are common with digital filters operating in the time domain. The disadvantage is that the frequency-based filters are computationally more demanding.

Different high- and low-pass filters were employed separately. When the analogue high-pass filter was set at 0.16 Hz, the low-pass digital filter was set at either 24, 12 or 6 Hz. When the low-pass analogue filter was set to 35 Hz, the high-pass digital filter was set at either 1, 2, 3 or 4 Hz.

## RESULTS

The number of trials that were available for averaging during SWS of course varied across subjects. The mean number of averaged deviant trials per subject within SWS was 361. Figure 1 illustrates the standard and deviant waveforms during wakefulness and SWS recorded prior to the application of the digital filters. When subjects were awake, N1 and P2 following the standard were difficult to detect. The deviant waveform consisted of a long-lasting negativity that is not

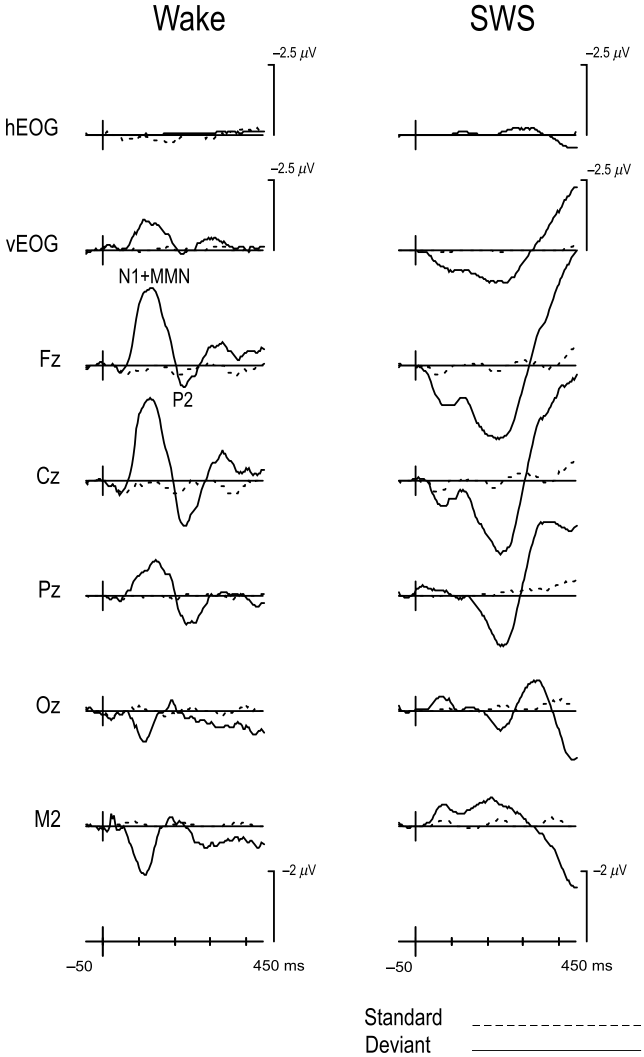


**Figure 1.** Grand average waveforms following the standard and deviant stimuli at all electrode sites during wakefulness and SWS prior to digital filtering. Analogue high- and low-pass filters were set at 0.16 and 35 Hz (3 dB attenuation), respectively.

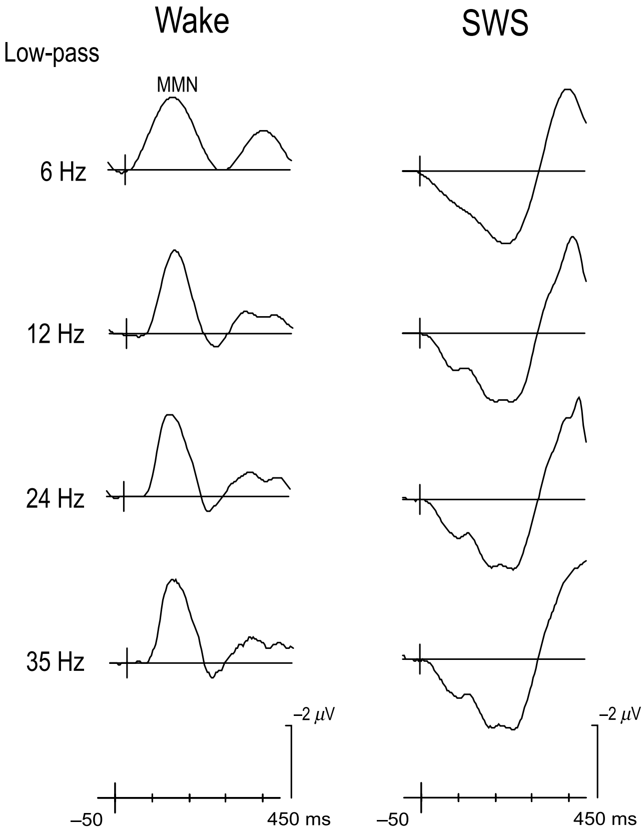


**Figure 2.** Low-pass filter effects on difference (deviant-standard) grand averages at the Fz site during wakefulness and SWS. High-pass filter was set at a constant 0.16 Hz.

apparent following the standard. This is the MMN. It is maximum over fronto-central areas of the scalp and is inverted in polarity at the mastoid. As may be observed, during SWS, this negativity is at or below baseline while the amplitude of the positive wave, P2, increases.

The effects of varying the low-pass filter on the difference waveforms of the wakefulness and SWS conditions are illustrated in Fig. 2. A clear MMN is evident during wakefulness. As the low-pass filter is changed from 35 to 6 Hz, there is a gradual 'smoothing' of the waveforms. It is only when the most extreme 6 Hz low-pass filter is used that the waking MMN becomes attenuated. The MMN is markedly attenuated during SWS. The low-pass filter settings did not, however, affect the MMN morphology.

The effects of applying the high-pass filter are illustrated in Fig. 3. During wakefulness, varying the high-pass filter from 0.16 to either 2 or 3 Hz resulted in a slight attenuation of the MMN. On the other hand, the 4 Hz filter caused a marked attenuation of the waking MMN. During SWS, the MMN was difficult to observe when the analogue filter was set to 0.16 Hz. Instead, a long-lasting positive wave emerged. The 1 Hz high-pass filter had no effect on this waveform. However, both the 2 and 3 Hz high-pass filters had a relatively large effect. The slow-wave now appeared as a negative-going potential that was maximum over fronto-central areas of the scalp. This MMN-like potential inverted in amplitude at the mastoid.

## DISCUSSION

In the waking state, the amplitude of N1–P2 was extremely small following presentation of the standard stimulus. The amplitude of N1–P2 is much reduced when stimuli are presented rapidly, as in the present study (see Näätänen and Picton 1987 for a review). A large amplitude MMN was observed during wakefulness. Low-pass filtering had a minimal effect on its morphology. Similarly, high-pass filtering below 3 Hz had a minimal effect. Only when a 4-Hz filter was applied was a sharp attenuation in the amplitude of the MMN observed.

During SWS, the deviant ERP was characterized by a long-lasting positivity that may have overlapped and summated

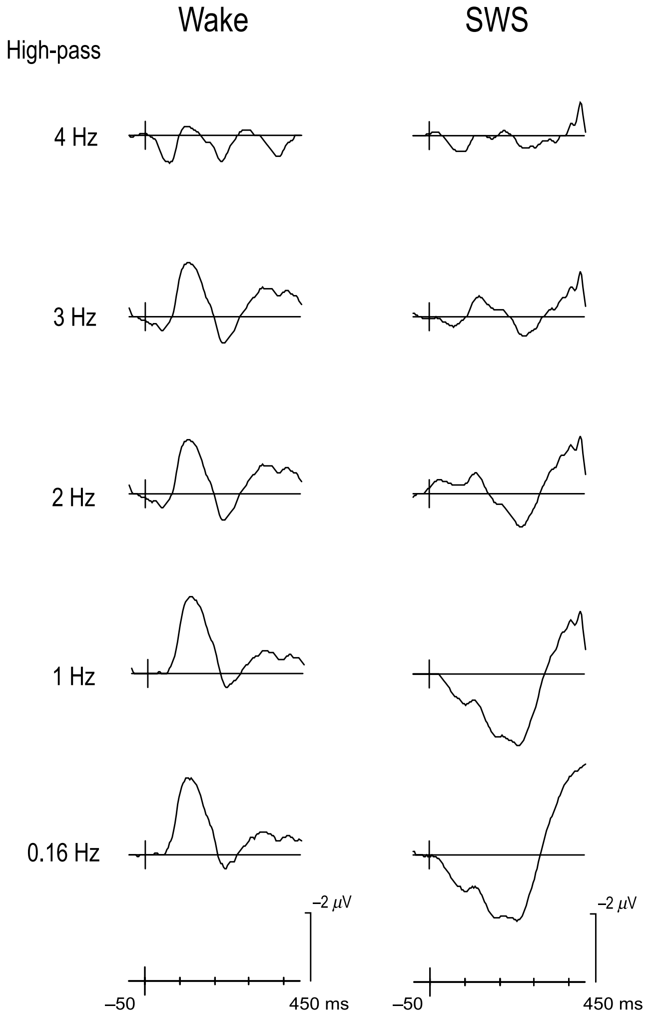


**Figure 3.** High-pass filter effects on difference (deviant–standard) grand averages at the Fz site during wakefulness and SWS. Low-pass filter was set at a constant 35 Hz.

with the negative-going MMN. A large number of studies have now reported a positive DC shift in ERPs during sleep onset and within non-REM (NREM) sleep (Atienza *et al.* 2000; Campbell *et al.* 1992; Harsh *et al.* 1994; Loewy *et al.* 1996; Sabri *et al.* 2000). This apparent positive-shift is probably because of the removal of the long-lasting waking negative wave from the waking ERP. This negative slow-wave overlaps and summates with the N1 and P2 components of the waking auditory ERP following the standard stimulus and also the MMN following the deviant. Campbell *et al.* (1992) suggested that this negative slow-wave reflects the additional processing that is received by attended channels during conscious states. During NREM sleep, the overall positive DC shift in ERPs is a reflection of the inhibition of sensory processing. This inhibition is greater for the processing of the rarely presented deviant stimuli.

The application of a low-pass filter had little effect on the MMN observed during SWS. The high-pass filter had a marked effect. Applying a 2–3 Hz high-pass filter will effectively remove the positive slow-wave, allowing the MMN to become visible. Caution should be exercised, however, when applying any filter to EEG data. All filters involve some compromise. The purpose of a filter is to attenuate unwanted 'noise'. At the same time, the actual signal should not be distorted. Distinguishing between signal and noise is not, however, always easy. This requires some knowledge of the frequency content of both the signal and noise. While the frequency spectrum of the MMN is relatively well understood in the waking state, it is possible that it is altered during unconscious states such as sleep. The type of filter that is employed is also not incidental. Digital filters operating in the time domain may introduce phase shifts in the signal of interest (for a discussion of this issue see Cook and Miller 1992). Digital filters operating in the frequency domain will overcome this problem, but are much more computationally complex. It is thus usually not possible to employ the frequency domain filters in real-time.

Few studies have been able to record the MMN during NREM sleep. The present study indicates that the usual 0.1–1 Hz high-pass filter setting will not allow the MMN to emerge during SWS. It may appear when the high-pass filter setting is set to 2–4 Hz. The 4 Hz setting will, however, distort the waking MMN. Still, if the high-pass filter is set to 3 Hz, the MMN will not necessarily be apparent in SWS (Loewy *et al.* 2000) or during deep coma (Fischer *et al.* 1999). Our purpose was not to resolve why the MMN was observed during SWS in the present study but not in others. Many factors could account for these differences. For example, the number of trials was much larger in the present study and the rate of stimulus presentation was much faster. What is clear, however, is that the high-pass filter setting used in most waking studies (0.1–1 Hz) will not allow the visualization of the MMN during sleep dominated by delta, slow-wave activity.

The present study demonstrates that low-pass filtering down to 12 Hz has a minimal impact on the waking or sleeping MMN amplitude. By contrast, the high-pass filter setting will have an effect. Increasing the high-pass setting to 2 or 3 Hz permits the visualization of the MMN. The 4 Hz filter markedly attenuates the amplitude of the waking MMN and thus is not recommended. A reasonable compromise might therefore be a 3 Hz high-pass filter.

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